

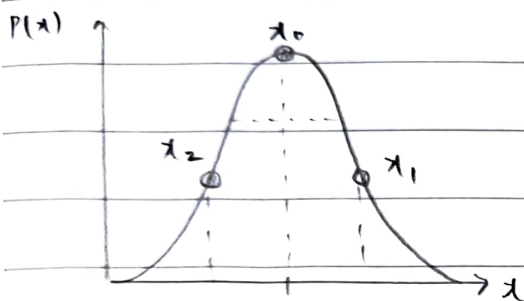
< The Unscented Kalman Filter (UKF) 7/15 >

UKF의 아이디어

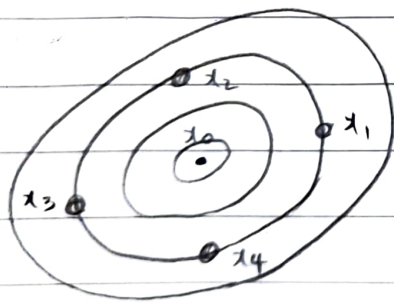
: 임의의 nonlinear function를 근사하는 것보다 확률 분포를 근사하는 것이 더 쉽다.

* choosing sigma points

: N -dimensional PDF $N(\mu_x, \Sigma_{xx})$ 에 대해 $2N+1$ 개의 sigma points 선택



→ 상태변수 하나에 대해서는 3개



→ 2개의 state vector에
대해서는 $2 \times 2 + 1 = 5$ 개

1) 공분산행렬에 대해 cholesky Decomposition을 적용

$$LL^T = \Sigma_{xx} \quad (L \text{ is lower triangular})$$

2) sigma points 계산

$$x_0 = \mu_x$$

$$x_i = \mu_x + \sqrt{N+K} \text{col}_i(L)$$

$$x_{i+N} = \mu_x - \sqrt{N+K} \text{col}_i(L)$$

$$(i=1 \dots N)$$

→ Gaussian PDFs에 대해서는

$$K = 9 - N \text{ 으로 권장.}$$

* Models

Nonlinear motion model : $x_k = f_{k-1}(x_{k-1}, u_{k-1}, w_{k-1})$

$$w_k \sim N(0, Q_k)$$

Nonlinear measurement model : $y_k = h_k(x_k, v_k)$

$$v_k \sim N(0, R_k)$$

* UKF steps

① Prediction

1) compute sigma points

$$\hat{L}_{k-1} \hat{L}_{k-1}^T = \hat{P}_{k-1}$$

$$\hat{x}_{k-1}^{(0)} = \hat{x}_{k-1} \text{ (mean)}$$

$$i = 1, 2, \dots, N$$

$$\hat{x}_{k-1}^{(i)} = \hat{x}_{k-1} + \sqrt{N+K} \text{col}_i(\hat{L}_{k-1})$$

$$\hat{x}_{k-1}^{(i+N)} = \hat{x}_{k-1} - \sqrt{N+K} \text{col}_i(\hat{L}_{k-1})$$

2) Propagate sigma points

$$\hat{x}_k^{(i)} = f_{k-1}(\hat{x}_{k-1}^{(i)}, u_{k-1}, 0)$$

$$i = 0, 1, \dots, 2N$$

3) compute predicted mean/covariance

$$\alpha^{(i)} = \begin{cases} \frac{K}{N+K} & (i=0) \\ \frac{1}{2} \frac{1}{N+K} & (\text{otherwise}) \end{cases}$$

$$\hat{x}_k = \sum_0^{2N} \alpha^{(i)} \hat{x}_k^{(i)}$$

$$\hat{P}_k = \sum_0^{2N} \alpha^{(i)} (\hat{x}_k^{(i)} - \hat{x}_k) (\hat{x}_k^{(i)} - \hat{x}_k)^T + Q_{k-1} \text{ additive process noise}$$

② Correction

1) Predict measurements from propagated sigma points.

$$\hat{y}_k^{(i)} = h_k(\hat{x}_k^{(i)}, 0) \quad i=0, 1, \dots, 2N$$

2) Estimate mean/covariance of predicted measurements.

$$\hat{y}_k = \sum_{i=0}^{2N} \alpha^{(i)} \hat{y}_k^{(i)} \quad (\text{mean})$$

$$P_y = \sum_{i=0}^{2N} \alpha^{(i)} (\hat{y}_k^{(i)} - \hat{y}_k)(\hat{y}_k^{(i)} - \hat{y}_k)^T + \textcircled{R_k}$$

additive measurement
noise

3) compute cross-covariance and Kalman Gain

$$P_{xy} = \sum_{i=0}^{2N} \alpha^{(i)} (\hat{x}_k^{(i)} - \hat{x}_k)(\hat{y}_k^{(i)} - \hat{y}_k)^T$$

$$K_k = P_{xy} P_y^{-1}$$

4) compute corrected mean/covariance

$$\hat{x}_k = \hat{x}_k + K_k(y_k - \hat{y}_k)$$

$$P_k = P_k - K_k P_y K_k^T$$

Jacobian 없음. Accuracy 향상, 연산속도는 느린 편

↳ UKF 사용을 최우선시 해라

* 각도를 다루는 λ, μ 에 대해서 무조건 잔차에 범위를 $(-\pi, \pi)$ 로 제한해야 함.

아니면 2의 2차가 358의 2차가 되어 K_k 라 곱해짐. K_k 는 삼각함수가 없어서
358은 2가 아니라 그 자체로 받아둬임.